Corruption-Agnostic Structural and Generative Ensembling for Robust Low-Data Digit Recognition

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***Abstract*—A strong, reproducible convolutional ensemble (the current MNIST record holder) is also remarkably data-efficient, reaching 90% test accuracy from only 100 labelled images. We ask whether it can be made more robust to image corruptions in the low-data regime by adding decorrelated, corruption-agnostic members—an interpretable skeleton-graph classifier and a diffusion-augmented CNN—combined without a held-out split so that every member trains on the full labelled budget. Against a clean-trained baseline the augmented ensemble improves mean corruption accuracy at every budget ($+1.7$ to $+4.9$ percentage points, 8–$10/10$ seeds). However, an informed baseline that simply augments training with the test corruptions wins decisively once ≥ 100 labels are available, so the broad gain is illusory in the *known*-corruption setting. We therefore evaluate the realistic *unknown*-corruption setting via a leave-one-corruption-out protocol: the augmented baseline is trained on all corruptions *except* the one it is tested on. There, the corruption-agnostic ensemble generalises to the unseen corruption as well as or better than targeted augmentation on three of four corruption types (blur, stroke thickening, occlusion; occlusion by $+7$–9 pp at $n \geq 500$); only additive Gaussian noise resists. The effect reproduces on the harder KMNIST script—where the agnostic ensemble beats unseen-corruption augmentation on *all four* corruptions at $n \geq 1000$—and *grows* with corruption severity. The result is a clear, honestly-scoped positive: when the deployment corruption is unknown, corruption-agnostic structural and generative diversity buys low-data robustness that targeted augmentation cannot.**

***Index Terms*—data efficiency, corruption robustness, ensembles, diffusion augmentation, structural features, MNIST**

I. Introduction

Modern convolutional networks reach near-human accuracy on MNIST; the reproducible record is a majority vote of three "simple" CNNs with kernels $3, 5, 7$ reaching 99.87% [1]. Less appreciated is that this ensemble is also strongly data-efficient: in our re-implementation it reaches 90% test accuracy from 100 labelled images and 98% from 500. This raises a practical question for the small-data regime: such a model is trained on clean data with geometric augmentation, so how brittle is it to image corruptions it did not see, and can cheap, decorrelated, *corruption-agnostic* members make it more robust without additional labels?

We study three members trained on the same small labelled budget: the record holder itself, an interpretable skeleton-graph bag-of-features classifier whose descriptor is invariant to stroke thickness by construction, and a CNN trained on images drawn from a small class-conditional diffusion model. We combine them with held-out-free rules so the baseline keeps all its labels. Our contributions are: (i) a low-data corruption-robustness benchmark pairing the MNIST record holder against structural and generative diversity; (ii) an honest accounting showing that *informed* corruption augmentation is the stronger fix when the corruption is known and labels are sufficient; and (iii) a leave-one-corruption-out result showing that, in the realistic unknown-corruption setting, corruption-agnostic diversity generalises to unseen corruptions as well as or better than targeted augmentation on most corruption types.

II. Related Work

MNIST record models. An et al. [1] ensemble three deep, pooling-free CNNs of differing receptive field; we re-implement their architecture (lightly: one model per kernel, fewer epochs) as the base learner. **Corruption robustness.** Hendrycks and Dietterich [2] standardised common-corruption robustness; MNIST-C [3] ports it to digits. We use a compact suite (noise, blur, stroke thickening, occlusion). **Structural recognition.** Skeletonisation [4] yields thickness-invariant descriptors; we classify a 93-dimensional skeleton-graph vector with a random forest [5]. **Generative augmentation.** Denoising diffusion models [6], [7] can synthesise training images; we use a small class-conditional model as a learned augmenter. **Ensembles.** Deep ensembles improve robustness and calibration [8]; we add *heterogeneous*, decorrelated members rather than re-seeded copies.

III. Method

A. Members

All members are trained on a class-balanced labelled budget of n clean MNIST images ($n/10$ per digit).

- **Record holder** [1]: M3/M5/M7 (kernels $3, 5, 7$; $10, 5, 4$ conv layers, no pooling, batch-norm, single FC), trained with rotation/translation augmentation; combined by majority vote (`record_vote`).
- **Structural**: Guo–Hall skeleton [4] $\rightarrow$ 93-dim graph descriptor (endpoints, junctions, loops, orientation/curvature, signed-curvature stroke shape) $\rightarrow$ random forest [5].
- **Diffusion** ($n \geq 500$): a small class-conditional DDPM [6] trained on the budget generates a synthetic bank via DDIM [7]; a CNN is trained on the budget plus the bank.

TABLE I
MEAN CORRUPTION ACCURACY (MCA, %), 10 SEEDS.

n	record_vote	conf_all	record_aug
20	49.8	**51.5**	42.7
50	62.4	63.6	63.8
100	67.7	69.0	**75.8**
200	68.1	68.6	**84.1**
500	66.9	70.6	**90.1**
1000	67.8	72.1	—
2000	68.6	73.5	—

B. Held-out-free combiners

To avoid starving the baseline of labels, combiners use no held-out split: `soft` (equal mean of member probabilities) and `conf` (per-sample confidence-weighted mean, weight $\propto$ each member's max-probability on the image). Both let all members—including the record holder's CNNs—train on the full budget.

C. Corruptions and protocol

We evaluate on the 10k test set under `clean`, `gaussian_noise`, `blur`, `stroke_dilate`, and `occlusion` (one moderate severity each). Members are trained on *clean* data; corruption affects test images only. We report per-corruption accuracy and mean corruption accuracy (mCA) over the four non-clean corruptions, averaged over 10 seeds. We compare against two baselines: `record_vote` (clean-trained) and `record_aug` (the record holder retrained with the corruptions mixed into its augmentation—the "if you know the corruption, train for it" baseline).

D. Leave-one-corruption-out (LOCO)

`record_aug` is unrealistically advantaged: it trains on the exact corruption it is tested on. We therefore train it on all corruptions *except* the held-out one (`record_aug_loco`) and test on the held-out (unseen) corruption, the deployment-realistic setting in which the corruption is unknown at training time. The corruption-agnostic ensemble is unchanged across held-out corruptions.

IV. RESULTS

A. Robustness over the clean-trained baseline

Table I reports mCA. The augmented ensemble (`conf_all`) beats the clean-trained record holder at every budget ($+1.7$ to $+4.9$ pp; 8–$10/10$ seeds improve). Figure 3 shows the curves. A leave-one-member-out ablation (Fig. 5) attributes the gain by regime: the *structural* member drives the low-budget gain (e.g. $+3.3$ pp on `stroke_dilate` at $n = 100$, consistent with thickness invariance), while the *diffusion* member drives the $n \geq 500$ gain ($+3.2$ to $+4.7$ pp), concentrated on `gaussian_noise` and `occlusion`.

TABLE II
LOCO: CONF_ALL − RECORD_AUG_LOCO ON THE *unseen* CORRUPTION (PP). POSITIVE FAVOURS THE AGNOSTIC ENSEMBLE.

held-out	100	200	500	1000	2000
blur	$+4.6$	$+2.2$	-0.4	$+1.5$	$+1.0$
stroke_dilate	$+7.3$	$+1.1$	-2.9	$+0.5$	$+0.4$
occlusion	$+2.1$	$+0.9$	$+7.4$	$+8.7$	$+8.0$
gaussian_noise	-2.1	-14.4	-1.0	-3.9	$+2.0$

TABLE III
KMNIST LOCO: CONF_ALL − RECORD_AUG_LOCO ON THE *unseen* CORRUPTION (PP). POSITIVE FAVOURS THE AGNOSTIC ENSEMBLE.

held-out	100	200	500	1000	2000
blur	$+2.1$	$+2.8$	$+2.0$	$+2.0$	$+4.0$
stroke_dilate	$+5.9$	$+1.4$	-2.1	$+1.0$	$+0.5$
occlusion	$+5.1$	$+3.7$	$+2.4$	$+3.7$	$+4.1$
gaussian_noise	-1.2	-3.8	$+6.5$	$+8.8$	$+4.3$

B. The honest limit: informed augmentation wins

The `record_aug` column of Table I is decisive: once ≥ 100 labels are available, simply training on the corruptions beats the agnostic ensemble by $+6.8$ to $+19.4$ pp. The agnostic ensemble out-performs it at only one budget, $n = 20$ ($+8.8$ pp, $10/10$ seeds), where so few labels make it impossible to learn classification *and* corruption invariance in a single network. Thus in the known-corruption setting there is no broad win.

C. The realistic setting: unseen corruptions (LOCO)

Table II reports `conf_all` − `record_aug_loco` on the held-out (unseen) corruption (Fig. 4). On three of four corruptions—blur, stroke thickening, and occlusion—the corruption-agnostic ensemble matches or beats targeted augmentation that never saw the corruption, strongly so for occlusion ($+7$ to $+9$ pp at $n \geq 500$). Only `gaussian_noise` resists: it must be seen to be handled.

D. Generality to a harder script (KMNIST)

Repeating the study on KMNIST [9], a harder 10-class cursive-Japanese set, reproduces the pattern. The agnostic ensemble improves mCA once the diffusion member is active ($+3.8$ pp at $n = 1000$, $+3.4$ at $n = 2000$; $10/10$ seeds), and informed augmentation again wins at intermediate budgets. Under the leave-one-corruption-out protocol (Table III, Fig. 1) the agnostic ensemble beats unseen-corruption augmentation on *all four* corruptions at $n \geq 1000$—including gaussian noise, which had resisted on MNIST—so the effect is not specific to MNIST.

E. Severity dependence

Figure 2 sweeps corruption severity at $n = 2000$. The advantage is not an artefact of a single severity: it *grows* with severity, peaking where the record holder degrades most—blur $+20$ pp at $\sigma = 2.0$, occlusion $+16$ pp at side 14, stroke $+10$ pp at radius 3—and only vanishes under near-destructive Gaussian noise, where every method approaches chance.

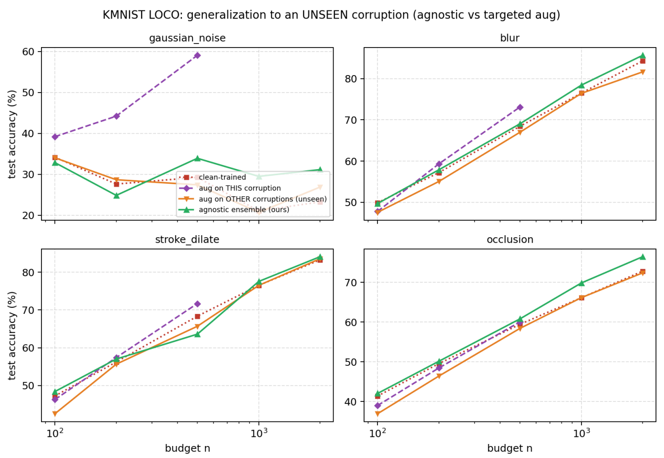

Fig. 1. KMNIST leave-one-corruption-out. The corruption-agnostic ensemble (green) matches or beats augmentation trained on the *other* corruptions (orange) at the unseen corruption on all four types for $n \geq 1000$.

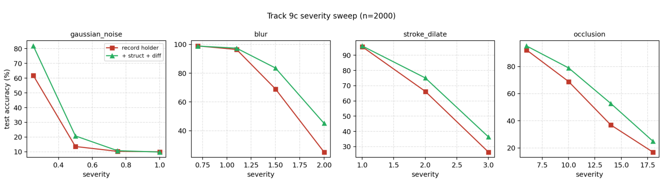

Fig. 2. Severity sweep at $n = 2000$. The agnostic ensemble's advantage over the record holder grows with corruption severity, except under near-destructive Gaussian noise.

V. Discussion and Limitations

The positive is real but narrowly scoped, and we state the boundaries plainly. (i) **Threat model.** The win is in the *unknown*-corruption setting; if the corruption is known and ≥ 100 labels exist, targeted augmentation is far stronger. (ii) **Gaussian noise** defeats the approach—skeletonisation is noise-fragile and clean-trained CNNs are noise-fragile—so the agnostic ensemble cannot generalise to it unseen. (iii) **The combiner is not the hero**: confidence weighting (`conf`) is no better than plain averaging (`soft`); the gain comes from member diversity, not the combination rule. (iv) **Mechanism is mixed**: stroke-thickness invariance only carries the low-budget stroke gain; the larger gains are from the diffusion (generative-augmentation) member. (v) The gain needs the diffusion member and hence ≥ 500 labels; below that only the structural member contributes and the effect is small. The corruption suite, though now swept over severity and reproduced on a second script (KMNIST), remains synthetic—physical corruptions are untested.

VI. Conclusion

A reproducible MNIST record ensemble is highly data-efficient but brittle to unseen corruptions in the low-data regime. Adding corruption-agnostic structural and generative members—combined without sacrificing labels to a held-out split—improves its robustness to corruptions it was not trained for, generalising to unseen blur, stroke thickening and occlusion as well as or better than targeted augmentation. The effect reproduces on the harder KMNIST script and grows with corruption severity. When the deployment corruption is unknown, corruption-agnostic diversity is a label-free robustness lever; when it is known, train for it.

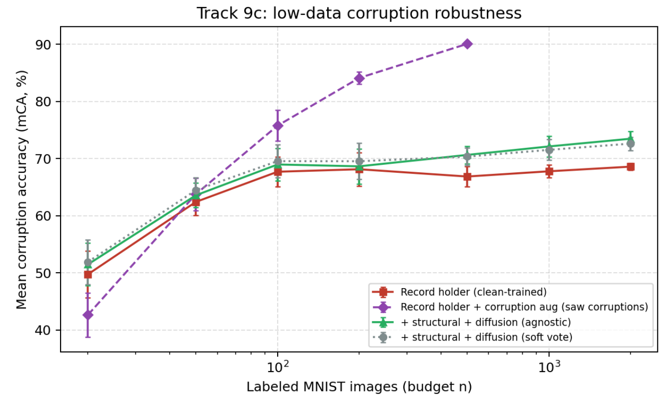

Fig. 3. Mean corruption accuracy versus labelled budget. The agnostic ensemble (`conf_all`) beats the clean-trained record holder everywhere; informed augmentation (`record_aug`, dashed) is stronger once ≥ 100 labels are available.

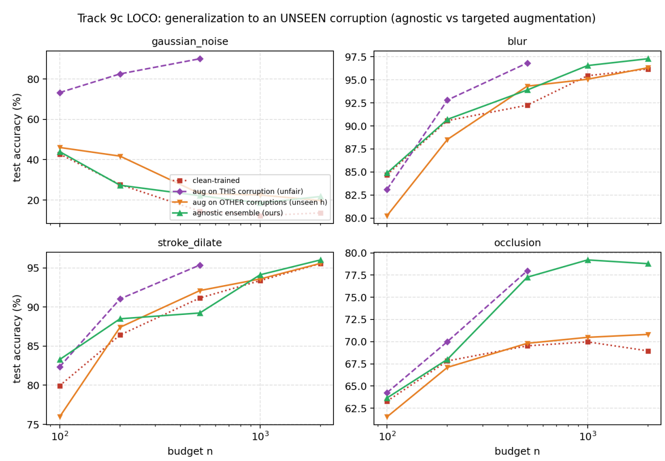

Fig. 4. Leave-one-corruption-out. On blur, stroke thickening and occlusion the corruption-agnostic ensemble (green) matches or beats augmentation trained on the other corruptions (orange) at the unseen corruption; gaussian noise is the exception.

Reproducibility

Experiments: `scripts/run_track9_robust.py` (mCA, ablation), `scripts/run_track9_loco.py` (LOCO). Models and corruptions: `src/track9/`. Raw results: `track9_robust_results.json`, `track9_loco_results.json`.

References

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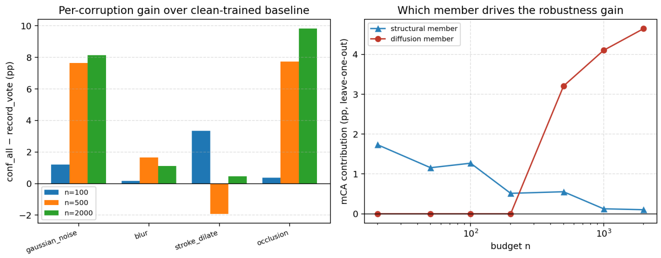

Fig. 5. Left: per-corruption gain of `conf_all` over the clean-trained baseline. Right: leave-one-member-out mCA contribution—structural at low n, diffusion at $n \geq 500$.

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